

4. GA 求解 0-1 背包问题 MATLAB 代码

func

```
function result = func4(f,C,W,V,afa)
fit = sum(f.*W);
TotalSize = sum(f.*C);
if TotalSize <= V
    fit = fit;
else
    fit = fit - afa * (TotalSize - V);
end
result = fit;
```

GA

```
%%遗传算法解决 0-1 背包问题
%%初始化参数
clear all; %清变量
close all; %清图
clc; %清屏
NP = 50; %种群规模
L = 10; %物品件数
Pc = 0.8; %交叉率
Pm = 0.05; %变异率
G = 100; %最大遗传代数
V = 300; %背包容量
C = [95,75,23,73,50,22,6,57,89,98]; %物品体积
W = [89,59,19,43,100,72,44,16,7,64]; %物品价值
afa = 2; %惩罚函数系数
f = randi([0,1],NP,L); %随机获得初始种群
%%遗传算法循环
for k = 1:G
    %%适应度计算
    for i = 1:NP
        Fit(i) = func4(f(i,:),C,W,V,afa);
    end
    maxFit = max(Fit); %最大值
    minFit = min(Fit); %最小值
    rr = find(Fit==maxFit);
    fBest = f(rr(1,1),:); %历代最优个体
    Fit = (Fit - minFit)/(maxFit - minFit); %归一化适应度值
    %%基于轮盘赌的复制操作
    sum_Fit = sum(Fit);
    fitvalue = Fit./sum_Fit;
    fitvalue = cumsum(fitvalue);
    ms = sort(rand(NP,1));
```

```

fiti = 1;
newi = 1;
while newi <= NP
    if (ms(newi)) < fitvalue(fiti)
        nf(newi,:) = f(fiti,:);
        newi = newi + 1;
    else
        fiti = fiti + 1;
    end
end
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%基于概率的交叉操作%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
for i = 1:2:NP
    p = rand;
    if p < Pc
        q = randi([0,1],1,L);
        for j = 1:L
            if q(j)==1;
                temp = nf(i + 1,j);
                nf(i + 1,j) = nf(i,j);
                nf(i,j) = temp;
            end
        end
    end
end
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%基于概率的变异操作%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
for m = 1:NP
    for n = 1:L
        r = rand(1,1);
        if r < Pm
            nf(m,n) = ~nf(m,n);
        end
    end
end
end
f = nf;
f(1,:) = fBest; %保留最优个体在新种群中
trace(k) = maxFit; %历代最优适应度
end
fBest; %最优个体
figure
plot(trace)
xlabel('迭代次数')
ylabel('目标函数值')
title('适应度进化曲线')

```